Telecommunication Systems
ICE 3109

Lecture 14
Integrated Services Digital Network (ISDN)

Integrated Services Digital Network

Introduction

Under the **existing telecommunication network scenario**, a **network** has been **constructed individually** for a **specific service**, as in a **telephone network**, a **packet network**, etc. (Fig 9.18(a)).

Accordingly, when using one or **more services**, the user **must** inconveniently **make** a subscription **contract**, lead in a **subscriber line** for **each service**, and have a **different number** for **each service**.

ISDN is intended to offer **different services** under the **same number (umbrella)**, employing a **subscriber cable** by **unifying individual networks** into a **digital network** (Fig 9.18(b)).

Overview

The **I**ntegrated **S**ervices **D**igital **N**etwork (**ISDN**) is a **set** of **digital transmission standards** which are used for **end-to-end digital connectivity**.

Integrated Services Digital Network

Introduction

Compared to the segregated telecommunication network architecture used earlier (Fig 9.18(a)), the common architecture of ISDN is shown in Fig. 9.18 (b).

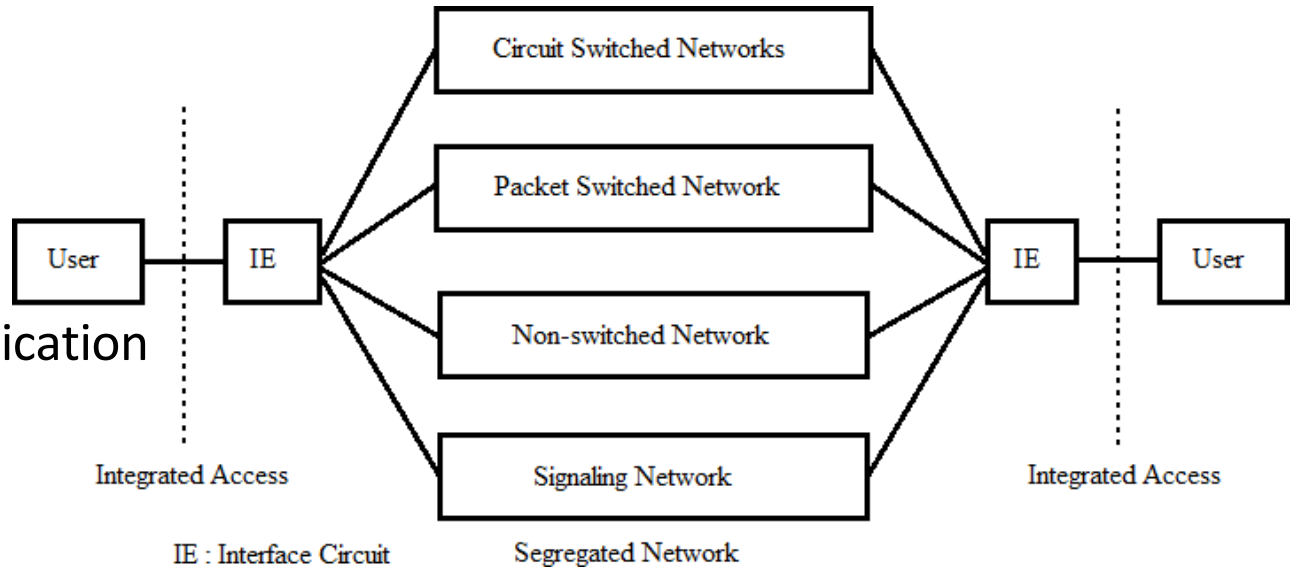


Fig. 9.18: (a) Telecommunication Architecture: (Segregated /separate)

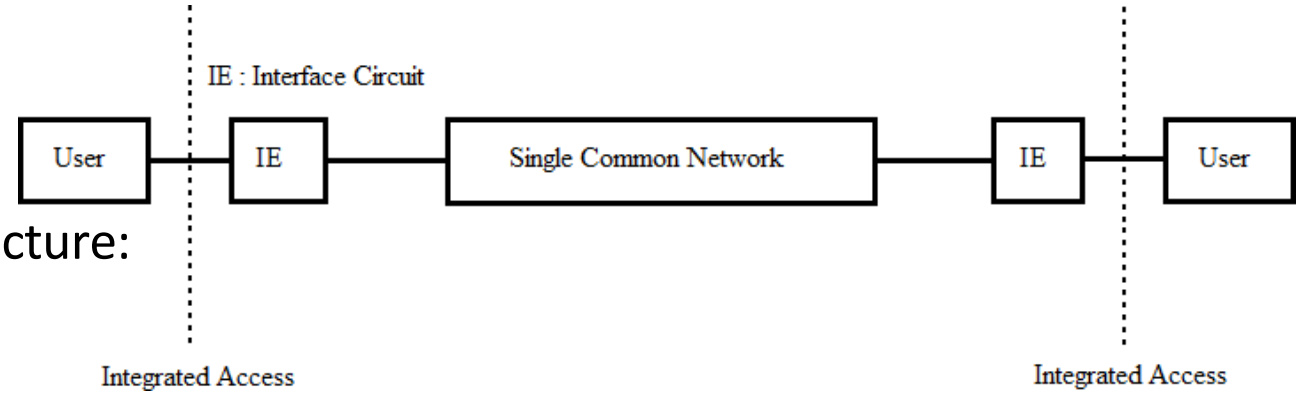


Fig. 9.18: (b) ISDN Architecture: Integrated (/common)

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Overview

In the past, video, audio, voice and data services required at least four separate networks (Fig 9.18(a)).

ISDN integrates all four over the same network (Fig 9.18(b)).

In general, ISDN networks extend from the local telephone exchange to the remote user and include all the telecommunications and switching equipments in between.

ISDN supports voice and data.

"Integrated Services" refers to its ability to **sustain numerous applications**. *"Digital Network"* is related to its **end-to-end digital connection**.

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History

ISDN is based on technology developed during the **1970's** designed to address the problem of **how to transport digital services** across a **telephony** infrastructure based on **copper** wiring **originally** intended to **carry analog signals only**.

To **meet** the **customer needs**, initially, the telephone companies created Integrated Digital Networks (**IDN**).

IDN is a combination of **networks available** for **different purposes**. **Access** to these **networks** is by **digital pipes** which are **time multiplexed channels** sharing very **high speed paths**.

By this network, customers can use their local loops to transmit both voice and data to their telephone company's central office.

The office then directs these calls to the appropriate digital networks via the digital pipes.

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History

ISDN integrates customer services with the IDN.

With ISDN, all customers will become digital rather than analog.

We know that a much more efficient means of transmitting data is via **digital transmissions**. Telephone companies across the country are installing digital central offices (DCO) equipped with digital switches.

The integrated services digital network will eventually replace all of the switches and circuitry now in the analog system (Fig. 9.13).

Standards

ISDN technology is standardized according to recommendations of the CCITT, now ITU, which describes the protocols and architectures to implement a world wide digital communication network. The CCITT was comprised to study groups (SGs). Each SG has its own area of expertise.

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ISDN Telecommunication Formation

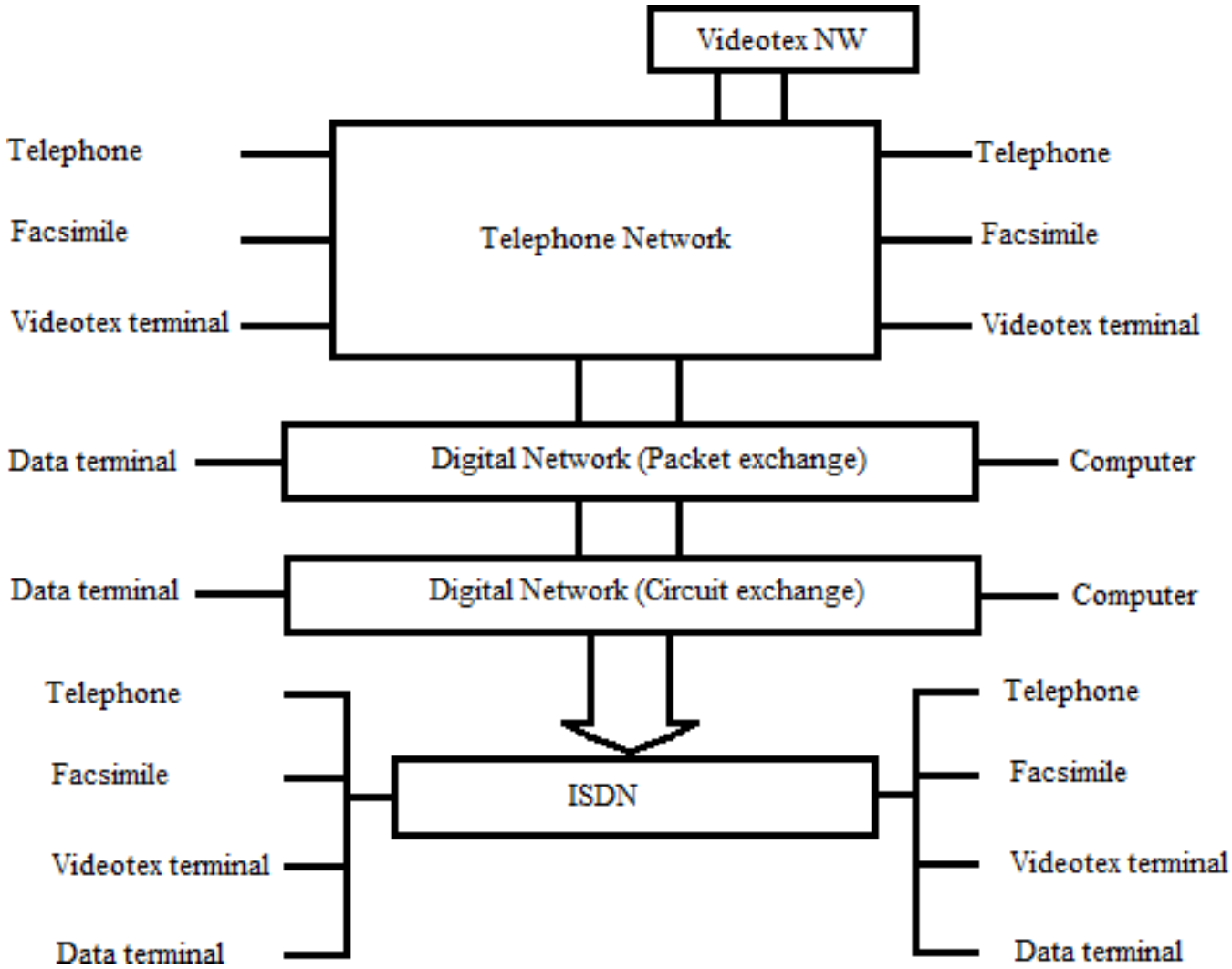


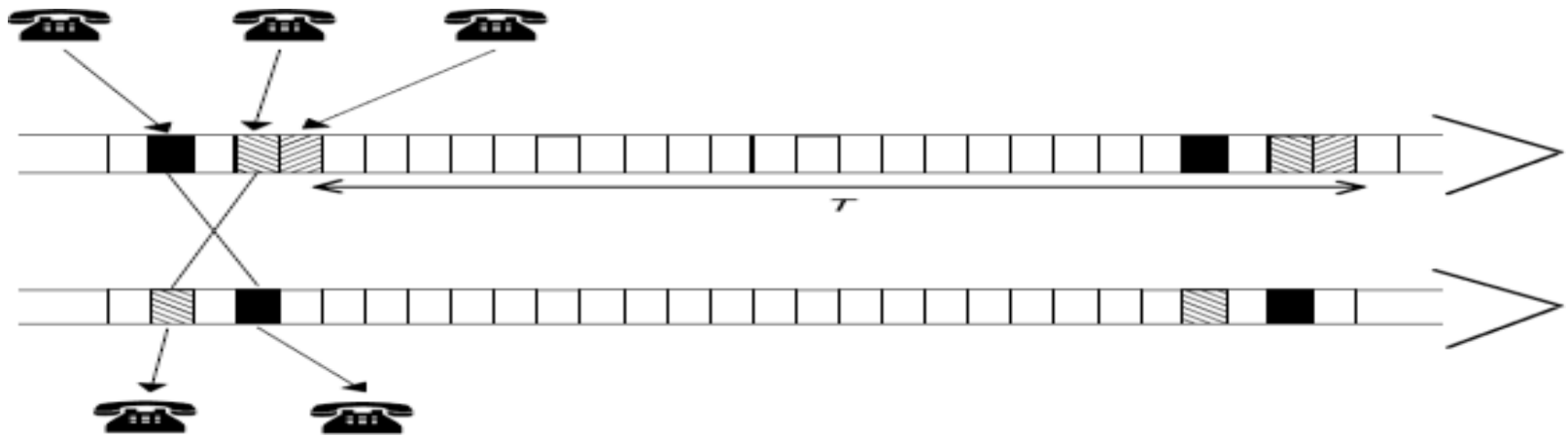
Fig. 9.13: ISDN Telecommunication Format.

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Introduction to ISDN: Digital Telephony

TDM Switching (Synchronous)

- PCM is to serve speech telephony, 64 kbps
- Not suitable for all the other services



ISDN: Summary

• ISDN is an inter LAN digital connectivity specification defined by ITU-T in 1988

• **ISDN** is a set of communications standards for simultaneous digital transmission of voice, video, data and other network services over the traditional circuits of PSTN

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Introduction to ISDN: Summary

- ISDN supports
  - ✓ circuit switching
  - ✓ packet switching and
  - ✓ non-switched (dedicated lines)
- Two principal standards:
  - ✓ High B/W for businesses
  - ✓ Low B/W for homes
- Supports multiple independent channels by TDM
- Supported Applications:
  - ✓ Digital voice (packet switched)
  - ✓ 64-kbit/s data, both circuit- and packet-switched
  - ✓ Telex/teletext
  - ✓ Facsimile
  - ✓ Slow-scan video

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ISDN Services

ISDN services generally fall into three categories. They are bearer services, teleservices and supplementary services.

Bearer services: ISDN works on the principle of transport services known as bearer services. The bearer service offers the capability to transport digital voice or non-voice services using this standard. The basic operation of the bearer service is the 64 kbps channel capacity.

Bearer services provide the means to transfer information (voice, data and video) between users. The network does not need to process the information. Bearer service belongs to the first three layers of OSI model. These services can be provided with circuit switched, packet switched, frame switched or cell switched networks.

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ISDN Services

Tele services: In this service, the network may change or process the contents. This service correspond to layers 4–7 of the OSI model. Tele services include telephony, telefax, videofax, telex and teleconferencing.

Supplementary services: It provides additional functionality to the bearer service and teleservices. Supplementary services include call waiting, Reverse charging, and message handling.

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Objective of ISDN

From the viewpoint of line utilization, individual networks constructed for individual services have low efficiency. Money is wasted by leading into users' premises the subscriber cables that occupy a larger part of the network cost for individual services.

In these days, the construction of digital network is made more economical than that of an analog network by the rapid advance of LSI technology and optical fiber transmission technology. This contributes greatly to the realization of ISDN.

The existing telephone network is formed on the analog technology most suitable for voice transmission. When considering, however, that the demands for such non-voice services as data communication and facsimile communication are increasing more than that for telephone service, it is desired that various services

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Objective of ISDN

can be offered in unified form by a user network interface that enables the use of many terminals simultaneously.

The objective of **ISDN** was introduced in **1979**. The concept is the **digital end to end connectivity** to support a **wide range of services**. The conceptual view of ISDN is shown in Fig. 9.1.

As shown in Fig. 9.13, ISDN is intended to offer diversified telecommunication services, including telephone and data communication services by a digital network.

Its objective is to enhance network efficiency, and to relieve subscribers from chores of individual contracts for each service, individual charges, individual numbers, and individual subscriber cables.

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Objective of ISDN

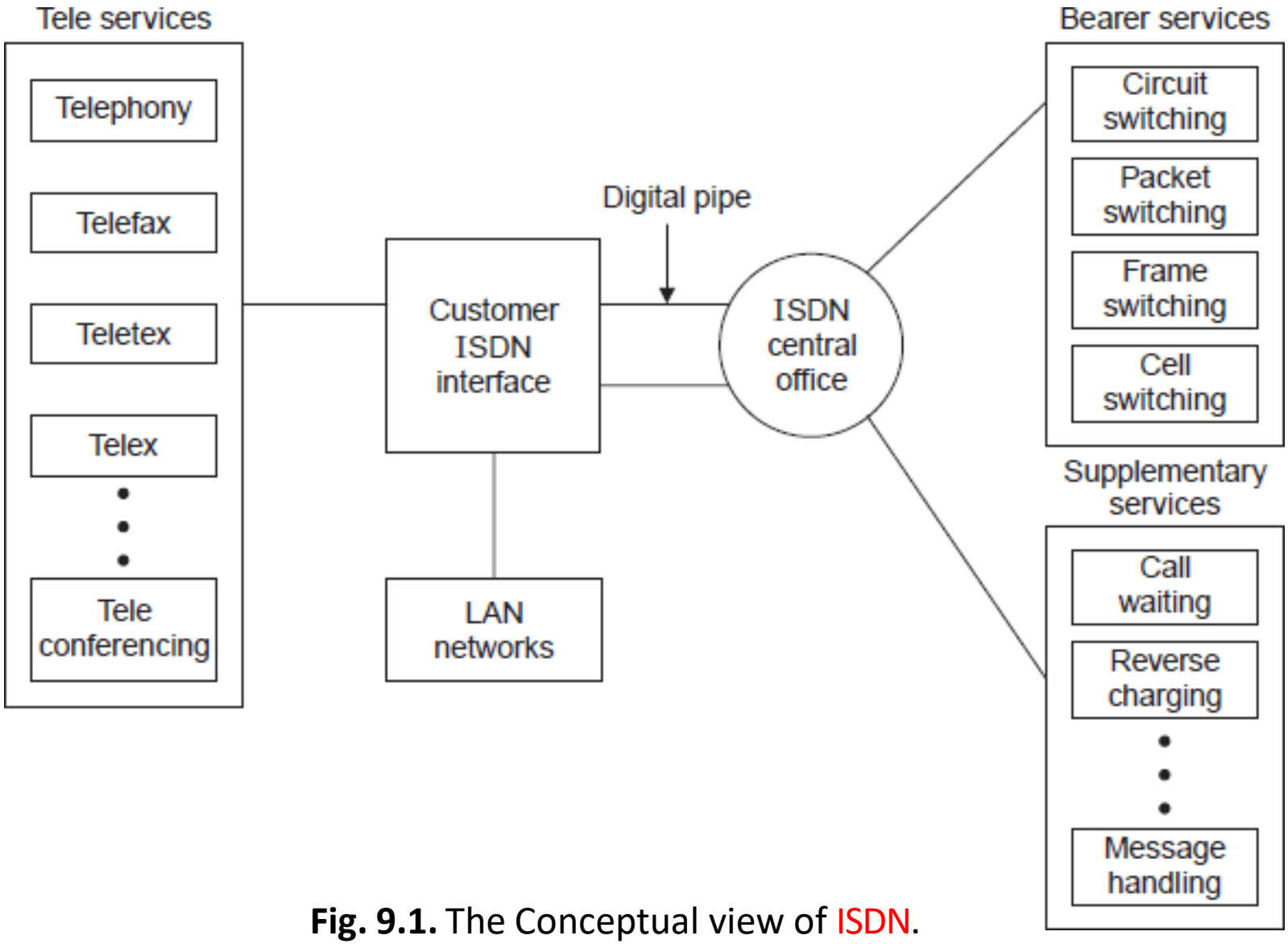
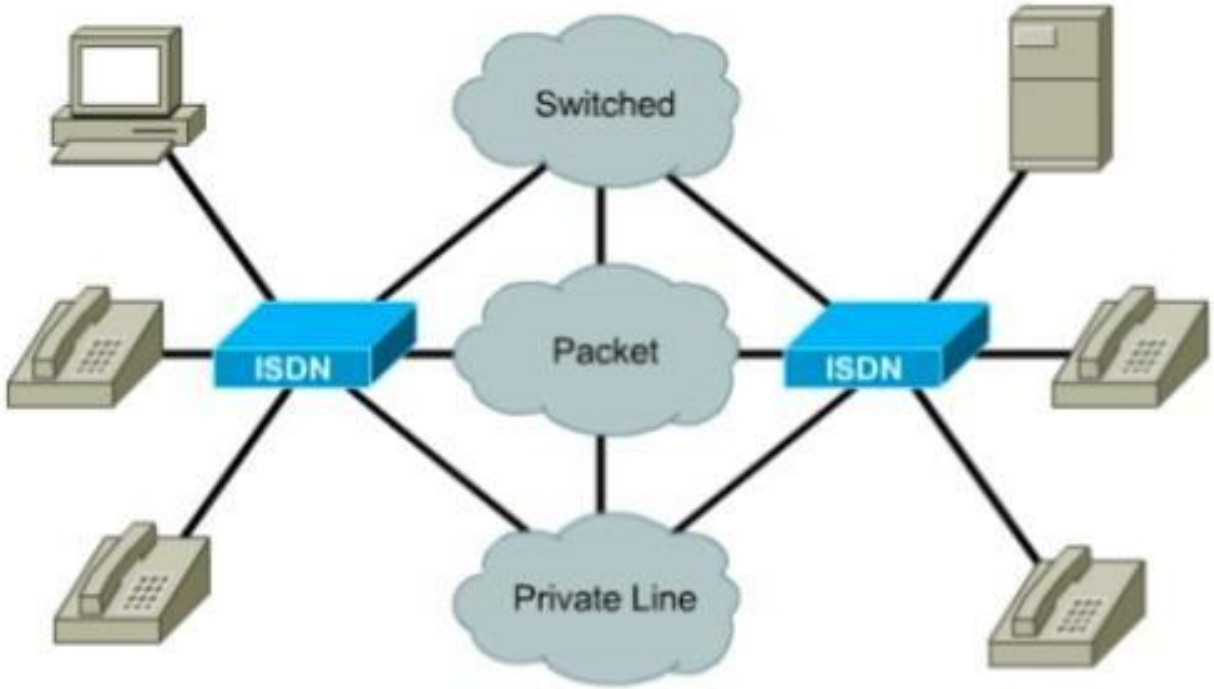


Fig. 9.1. The Conceptual view of ISDN.

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ISDN Network Connection





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ISDN Network Connection

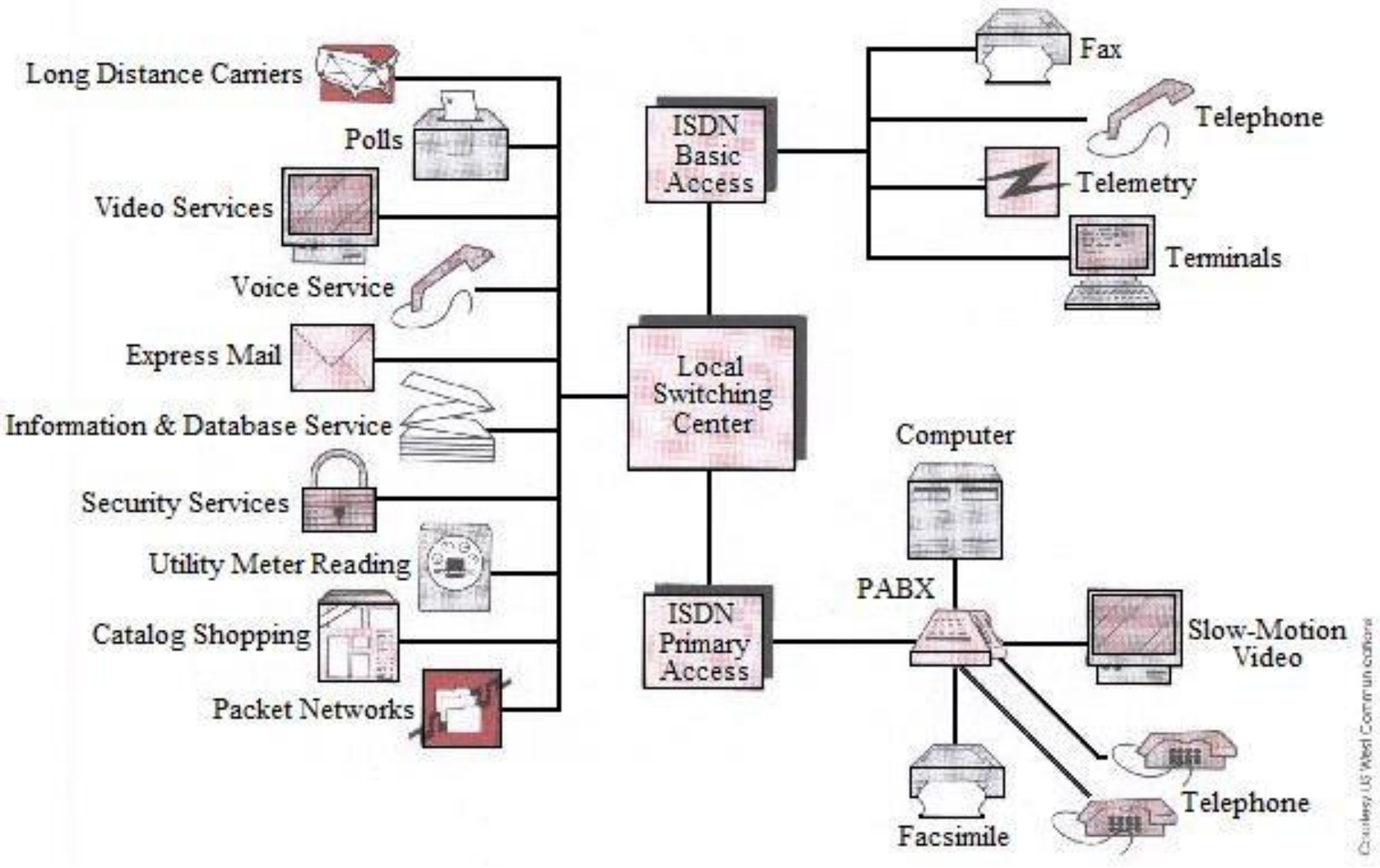


Fig. 9.10: ISDN Network Connection (Local Switching)

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Features of ISDN

1. Service Integration: In ISDN, multiple communication services (telephone, data communication, facsimile) and multiple communication modes (circuit-switching, packet switching) can be offered by one network.

2. International Standardization: In ISDN, international standardization is required so that diversified telecommunication services may be received by any user terminal throughout the world.

Unifying the technical standards of ISDN permits the international liberalization of telecommunication, the promotion of internationalization, and the wider use of terminal equipment.

The technical standards of ISDN are recommended by ITU-T.

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Features of ISDN

For connection between terminals and a network, the connecting conditions must be clarified because services other than telephone services are realized by different terminals.

The user network interface specifies these connecting conditions. The user network interface specifies connecting points between terminals and a network, and communication protocols at these connecting points.

The user network interface is also called the I-series interface because it is specified in the I-series recommendation of the ITU-T recommendations.



Fig. 9.14: Concept of User Network Interface.

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Features of ISDN

3. Simultaneous Use of Multiple Channels: In ISDN, with a contract for one subscriber, communication can be made different points by using more than one channel. For example, while a facsimile message is being transmitted from A to B, a conversation can be conducted with C.

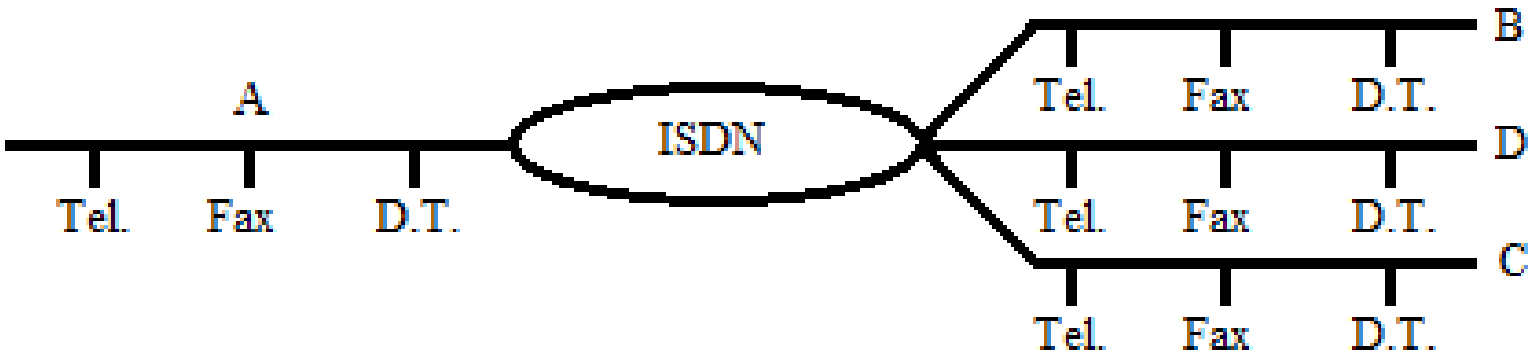


Fig. 9.15: Example of Simultaneous Telecommunications.

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Features of ISDN

4. Separation of Information and Signaling Channels: By separating the information channels and signaling channels, signals can be exchanged during communication, thereby permitting the provision for diversified services.

(a) Serviced that can be offered by transmitting signals during communication:

- Connected line identification presentation
- Moving equipment during communication

(b) Services that can be offered by transmitting/receiving a large quantity of signals:

- Calling line identification presentation
- Advice of charge
- User-to-user signaling
- Connected line identification presentation

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ISDN Channels

ISDN is an **all-digital network**. It consists of **three** types of **communications channels**. They are:

1. Bearer channel (**B channel**)
2. Data channel (**D channel**), and
3. Hybrid channels (**H channel**).

B Channel - 64 kbps
D Channel - 16 or 64 kbps
H0 Channel - 384 kbps
H11 Channel - 1536 kbps
H12 Channel - 1920 kbps

These three **ISDN** channels are described next.

B channel: B channels are logical digital ‘‘pipes’’ which exist on a single ISDN line. B channel carry data and services at 64 kbps. It carries data in full duplex mode. Each B channel provides a 64 kbps clear channel, **clear** meaning that the **entire bandwidth** is **available** for **data**, B channels typically form **circuit switched connections**.

B channel connection is an **end-to-end physical circuit** that is **temporarily dedicated** to **transferring data** between **two devices**[2][1].

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ISDN Channels

The circuit switched nature of B channel connections; combined with their reliability and relatively high bandwidth makes ISDN suitable for a range of applications including voice, video, fax and data. B channels are normally used for on-demand connection.

As B channel operation is based on circuit switching, it can be configured as semi permanent or ‘‘nailed up’’ connections.

D channel: D channel can be either 16 or 64 kbps, depending on the needs of the user. The primary function of the D channel is to carry control signaling and administrative information for B channels to set up and tear down the calls.

The D channel uses packet switched connection. The packet switched connections are best adapted to the intermittent but latency sensitive nature of signaling traffic, accounting for the highly reduced call setup time of 1 to 2 seconds on ISDN calls.

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ISDN Channels

Unlike the B-channel, which can function as a simple 'pipe', the D channel is associated with higher level protocols at layers 2 and 3 of OSI model which form the packet switched connections.

The D channel provides the signaling information that is required for caller identification. It also includes low-rate data transfer and applications such as telemetry and alarm transmission.

H channels: H channels are suitable for high data rate applications such as video, teleconferencing and so on. Table 9.2 gives ISDN channels and their specifications.

Channel	Bit rate (kbps)	Interface	Purpose
B	64	BRI	Bearer services
H0	384	PRI	6 B channels
H11	1536	PRI	24 B channels
H12	1920	PRI	30 B channels
D	16	BRI	Administrative and control signalling
D	64	PRI	,,

Table 9.2: ISDN channel specifications

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ISDN User Channels: Summary

- Supports multiple channels interleaved by TDM
- Three types channels: B, D and H
  - **Bearer (B)-channel**
  - ✓ Basic user channel
  - ✓ 64 kbps
  - ✓ Serve all types of traffic

  - **Data (D)-channel**
  - ✓ 16 or 64 kbps
  - ✓ 16 kbps is called basic rate and 64 kbps is call primary rate
  - ✓ It is a common channel for providing control signals to all other channels.

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ISDN User Channels

- **Hybrid (H)-channels : H0 and H1**

- Supports higher speed services
- H0 channel: 384 kbps
- H1 channels: 1.472 Mbps (H10), 1.536 Mbps (H11) and 1.920 Mbps (H12)
- Carry a variety of user information streams, e.g.,
  - Fast facsimile
  - Video, such as video conferencing
  - High-speed data
  - High-quality audio or sound program channel
  - Packet-switched information

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Types of ISDN and Their Use

Two types of ISDN: Narrow-band ISDN (**N-ISDN**) and Broad-band ISDN (**B-ISDN**).

N-ISDN: In N-ISDN, there are two interfaces depending on the information volume used for communication. The **two user interfaces** of **N-ISDN** are: Basic Rate Interface (**BRI**) and Primary Rate Interface (**PRI**).

Basic Rate Interface (BRI) – Intended for Home Use: BRI is made up of two full duplex 64 kbps B-channels (Bearer channels) and one full duplex 16 kbps D channel. Therefore the total rate is **2B + D**. B channels are 64 kbps and can be used for voice and data communications. The D channel is 16 kbps and is used for call initialization and signaling connections.

For this service, existing subscriber cables can be used as they are.

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N-ISDN: Basic Rate Interface/Access

Fig. 9.2 shows the ISDN BRI.

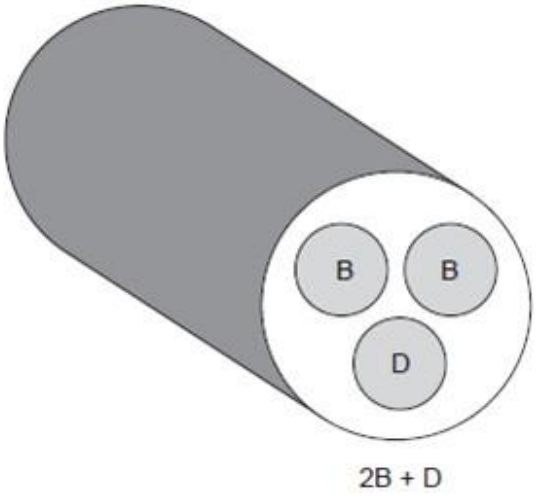


Fig. 9.2: ISDN BRI.

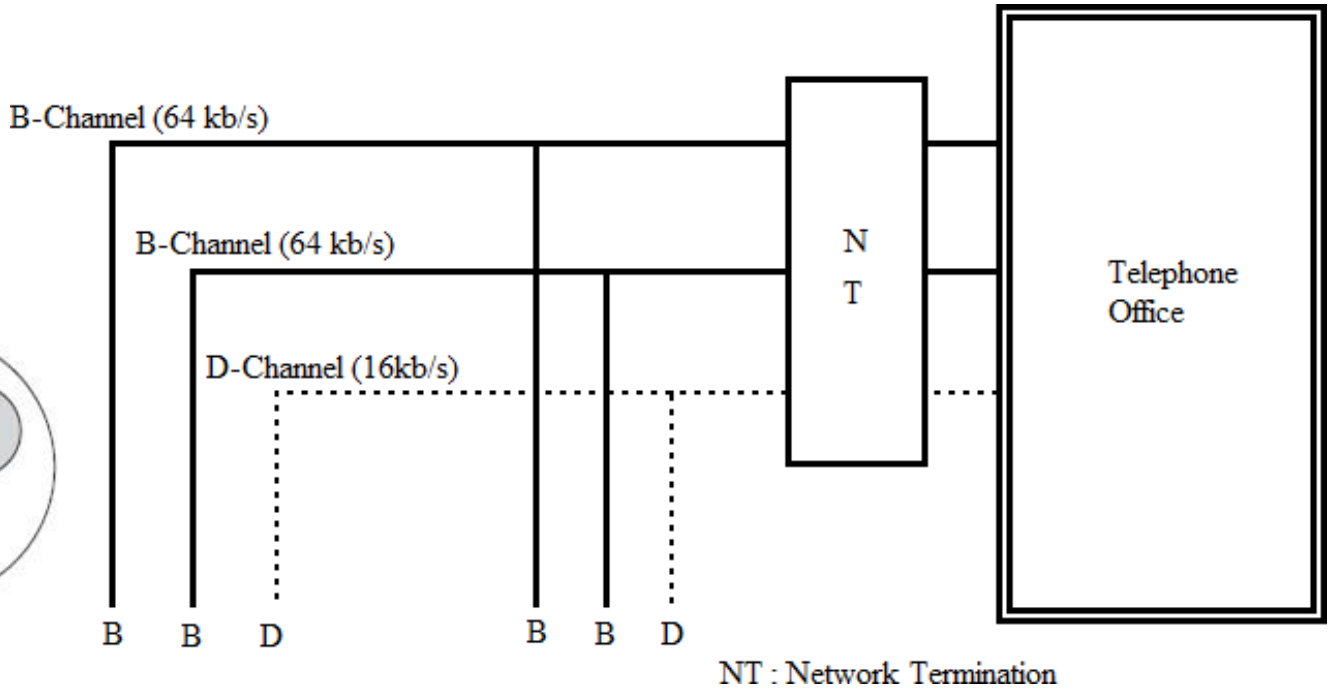
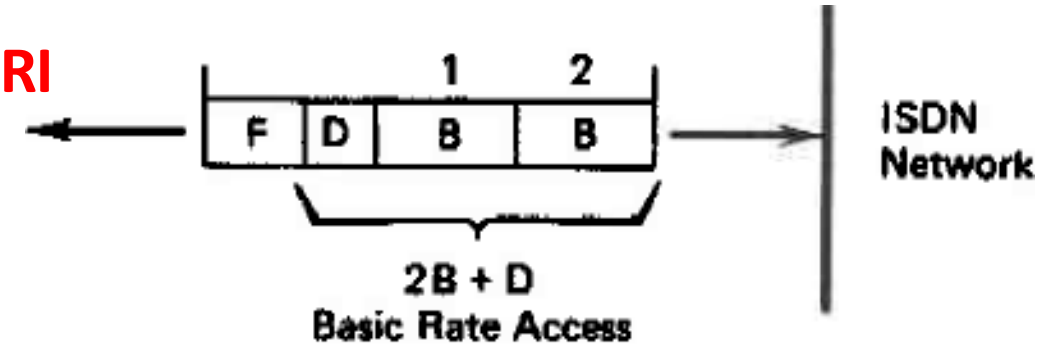


Fig. 9.16: The Basic Rate Interface.

ISDN lines often are 3 to 4 times **faster** than **modem transmission**. Furthermore, ISDN **lines** are very **clean**, so error correction does not consume a significant amount of transmission time.

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N-ISDN: BRI



The **B-channels** may be used **independently** (i.e., two different simultaneous connections).

BRI and ISP: With a **regular modem connection**, one **cannot call out while** using the **line** for **data**, and incoming calls either get a busy signal or disrupt the modem connection.

The **ISDN** is **integrated** with the **regular telephone system**. Using one ISDN B channel to link to an **I**nternet **S**ervice **P**rovider (**ISP**) still leaves **one** ISDN **B channel free** for **telephone calls**.

Alternatively, some Internet Service Providers (**ISP**s) allow one to **connect** to them with **both B channels simultaneously**, giving a speed of **128 kbps**.

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Basic Rate Interface (BRI) (2B +D)

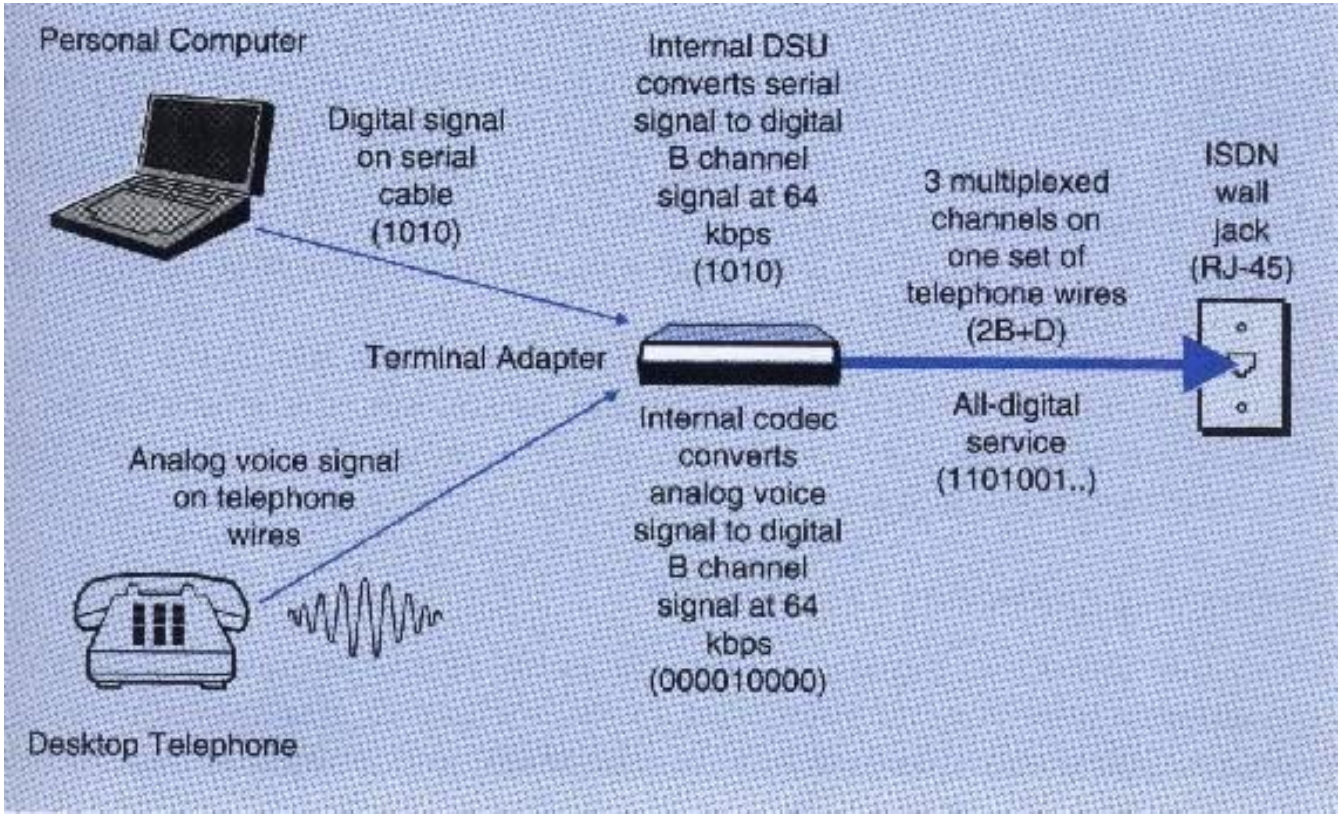


Figure: **ISDN Service**: ISDN's Basic Rate Interface (BRI) brings three transmission channels to the user's desktop. There are two 64 kbps B channels and one 16 kbps D channel.

These **three channels** are **multiplexed** onto **a single wire bundle** to the desktop.

Computers need **terminal adapters (TA)** or internal **DSU**s to communicate over the **ISDN**.

Analog telephones need **codecs** to translate their analog signals into **digital ISDN signals**.

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N-ISDN: BRI

BRI is the most appropriate type of **ISDN service** for computer connections and individual use. Of three digital channels **(2B + D)**, of each of the channels can be used simultaneously. Thus a subscriber can perform several communications tasks at the same time.

BRI is designed to carry the most data possible to the home through **existing copper phone lines**.

For every ISDN line, Network Terminator type 1 (NT 1) and a power supply are required. A special terminal adapter can combine the two B-channels to create a 128 kbps channel which can then be connected to a computer. The NT1 connected to ISDN line acts as a multiplexer and demultiplexer.

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N-ISDN: BRI

Fig. 9.3 illustrates the BRI concept.

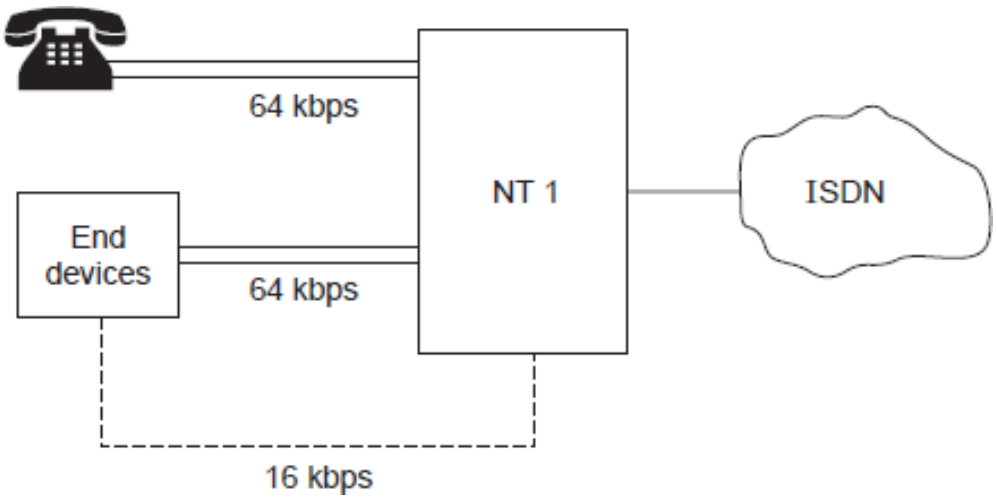


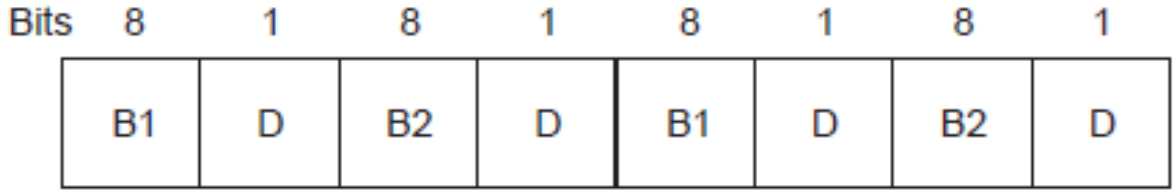
Fig. 9.3: BRI concept.

This is the basic service that **ISDN** brings to each desktop including three transmission channels. There are **2 B channels** operating at **64kbps**. There is also a **D channel** operating at **16kbps**. These **3** channels are multiplexed (mixed) over a single set of wires leading to the desktop. There is **no need** to have **3** separate **sets** of **wires**.

ISDN BRI Frame Format: The frame format for **BRI** is shown in Fig. 9.4. The size of a **frame** is **48 bits**.

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N-ISDN: BRI Frame Format



B1 : 16 bits B2 : 16 bits D : 4 bits **overhead : 12 bits (not shown)**
BR1 = 4000 frames/sec = 4000 × 48 bits/frame = 192 kbps

Fig. 9.4: BRI frame format.

Each **B channel is sampled two times** during each frame (**8 bits** per **sample**). **D channel is sampled four times** during each frame (**one bit** per **sample**). The overheads are not shown.

The reason for **interleaved sampling** is to **create long frame**. This is useful in many cases, for example **the size of the BRI frame** matches for a **data portion of an ATM cell**.

BRI functionality is **attainable without** any **modification** to the **existing telephony infrastructure**. Telephone **companies** must **change** the **signaling** on the **local loop** to **support ISDN**, but **no physical modifications** are **required**.

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N-ISDN: PRI

Primary Rate Interface (PRI) – Intended for Office Use: PRI in North America has **23 B** channels and one **64 Kbps D channel** or the total rate is **23 B + D**, having a total bandwidth of **1.544 Mbps**. (including **8 kbps** of **overhead**).

PRI in rest of the world uses **30 B** channels and **one D** channel or **30 B + D** with total rate of **2.048 Mbps**. The number of B channels is limited by the size of the standard trunk line used in the region. Fig. 9.5 shows the ISDN **PRI interface**.

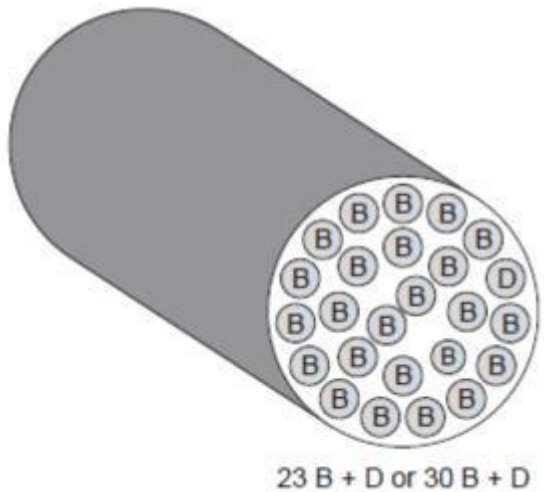
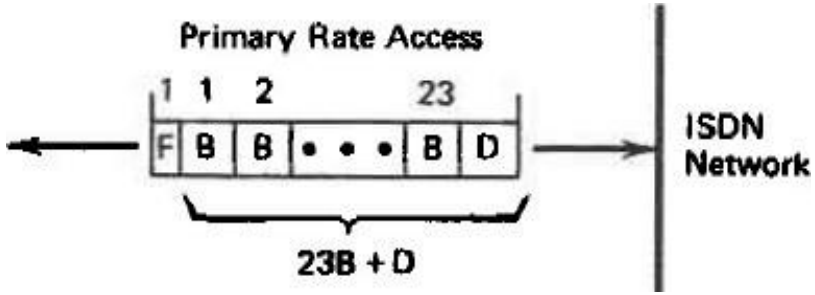


Fig. 9.5: ISDN PRI.

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N-ISDN: Primary Rate Interface/Access

- Intended for customers with greater capacity requirements
- The primary rate interface (PRI) structures are composed of *n* B-channels and one D-channel
  - D-channel (control and signaling) for PRI is 64 kbps
  - There are two primary data rates:
    - 23B + D + 8kbps overhead: 1.544Mbps (US, Canada, Japan)
    - 30B + D + 64kbps overhead: 2.048Mbps (Europe)
- May also support H channels
  - **PRI H0:** H0-channel structures are composed of H0 channels with or without a D-channel
    - 3H0+D or 4H0 + overhead = 1.544 Mbps
    - 5H0+D + overhead = 2.048 Mbps



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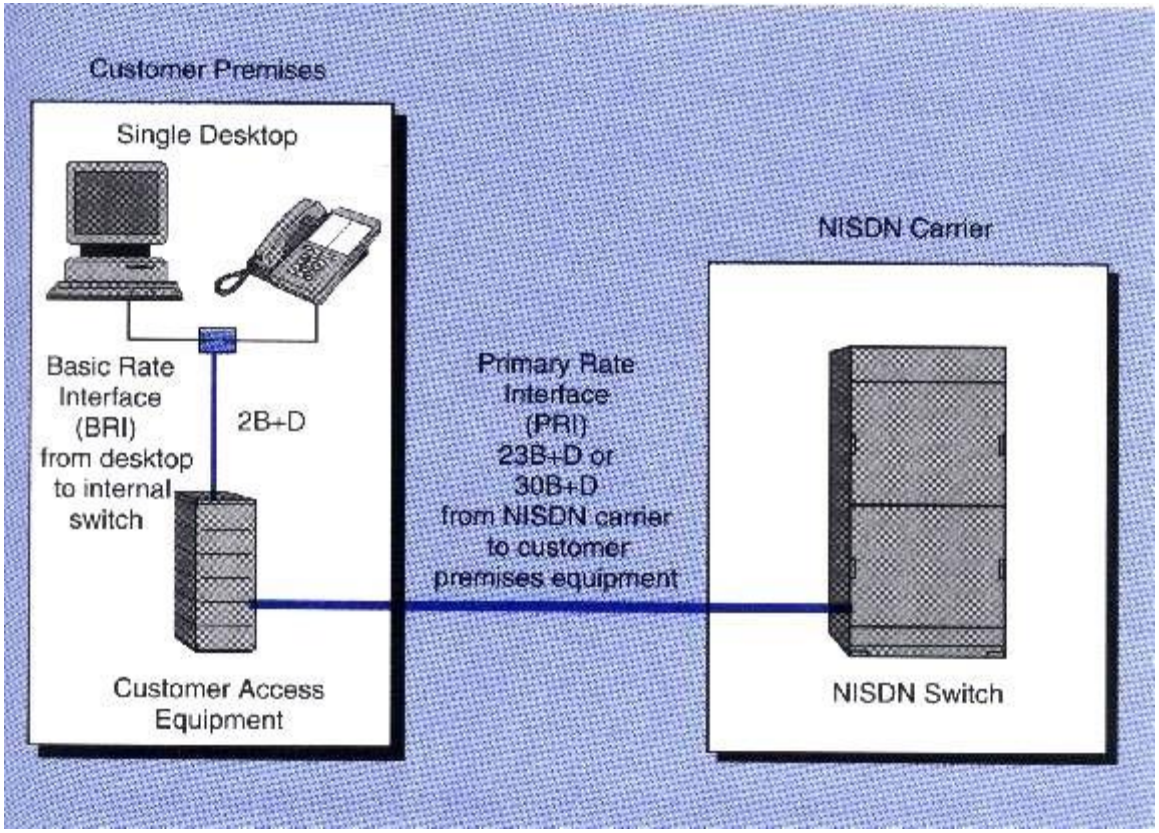
N-ISDN: PRI

Primary Rate Access Interface (**PRI**) may be defined around B, H0, H11, and H12 channels:

1544 kbps	2048 kbps
23B+D	30B+D
3H0+D	5H0+D
H11	H11+D

Figure: Narrowband ISDN

Narrowband ISDN (NISDN) offers a Basic Rate Interface (BRI) to each desktop. This interface multiplexes three signals. One is a 16 kbps D control channel. The other two B channels to two stations, say, using one for voice and one for data.



The connection from the customer premises to the ISDN carrier follows the Primary Rate Interface (PRI), which operates at 1.544 Mbps or 2.048 Mbps.

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N-ISDN: PRI

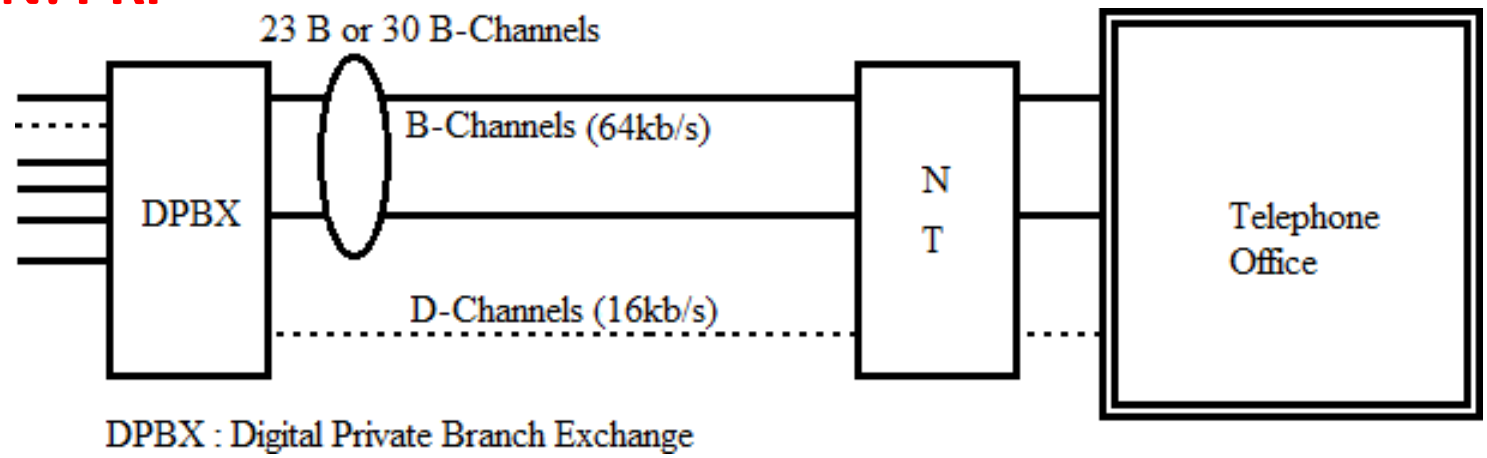


Fig. 9.17: The Primary Rate Interface.

A group consisting of more than one 64kbit/s information channel can also be used as H0 (384kbit/s: 64x6) and H11 (1536kbit/s: 64 x 24) or H12 (1920kbit/s: 64 x 30) information channels.

Above that H21 provides 32Mbps (512 B channels); H22 provides 44Mbps (690 B channels) etc.

Note: *The existence of the two types of (23B+D) and (30B+D) is because of a difference between the digital hierarchy adopted in Japan, the United States and Canada, and the Hierarchy adopted in Europe.*

Primary User Interface

- **PRI H1:**
  - One H11 + Overhead = 1.544 Mbps (1.536 Mbps H11-channel structure is composed of 24 B channels)
  - One H12 + Overhead = 2.048 Mbps (1.920 Mbps H12 structure is composed of 30 B channels)
- **PRI of Mixture of B and H0:**
  - Any combination of B and H0 channels and with/without a D-channel
  - e.g., 3H0+5B+D or 3H0+6B + overhead = 1.544 Mbps
- Signalling for the H channels, if required, is carried on the D-channel of another interface structure.

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N-ISDN: PRI

Unlike BRI, PRI does not support a bus configuration and only one device can be connected to a PRI line. A PBX, however, can reallocate ISDN PRI resources on to multiple BRI **buses**. Fig. 9.6 shows the ISDN PRI concept.

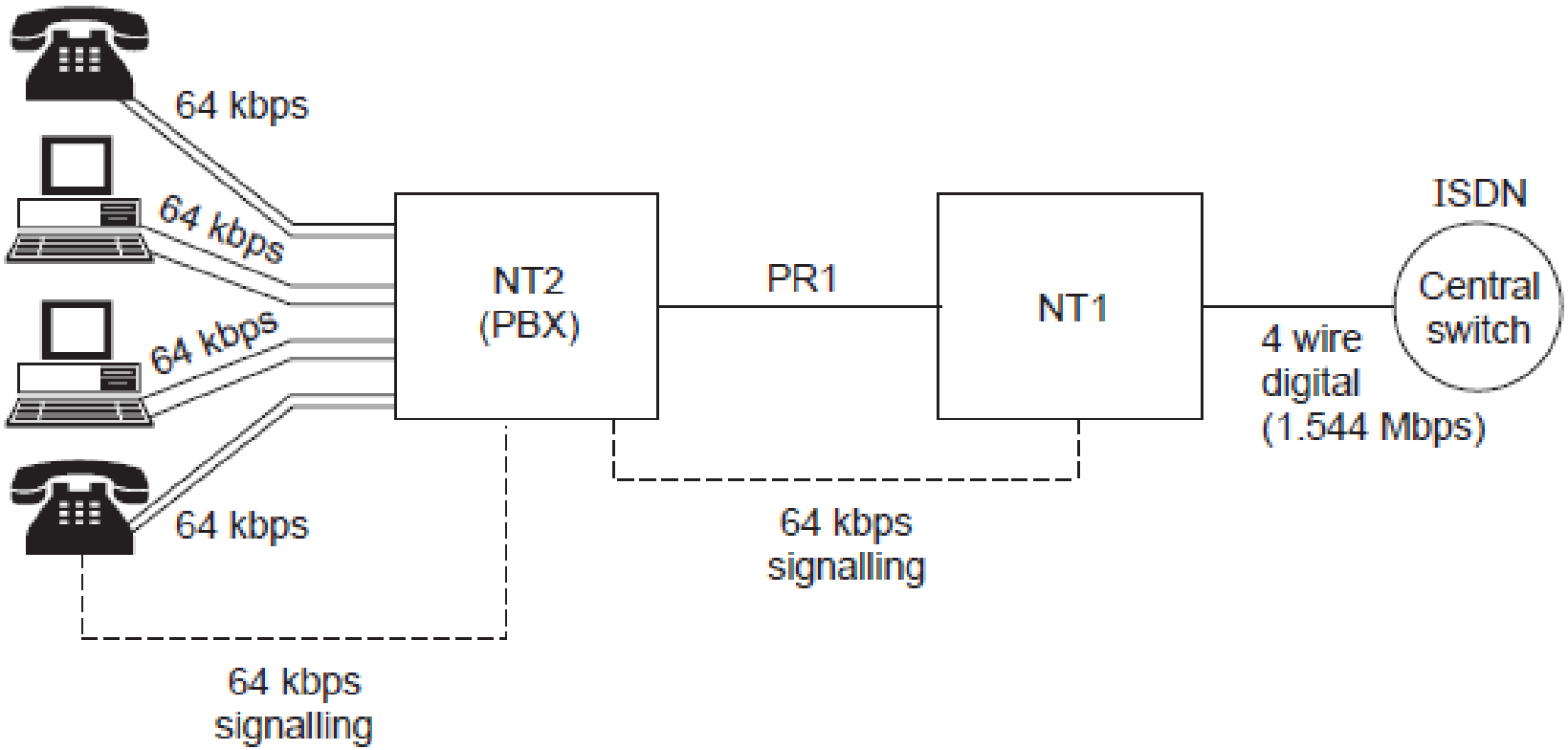


Fig. 9.6: ISDN PRI Concept.

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N-ISDN: PRI

A single PRI connection is usually much less expensive than obtaining the equivalent number of B channels through multiple BRI connections. The **1.544 Mbps** of a **PRI** can be divided up in many ways to meet the requirements of many users. This indicates combination of 64 kbps B channels or all capacity of B channels (In LAN to LAN case) can be used.

Application of PRI: NT 2

Network Terminator 2 (NT2): An application of the PRI is to connect two central switches together to use them as a **T1** link (Fig. 9.6). The devices which handle switching and multiplexing (such as PBX) are called network terminator 2 (**NT2**).

ISDN **PRI** can **connect** the **customer directly** using an **NT2 device**, while ISDN **BRI requires** an **NT1 device**. **NT2** performs **functions** at the **physical**, **data link** and **network layers** of the **OSI model**.

Integrated Services Digital Network

N-ISDN: PRI

An NT2 provides intermediate signal processing between the data generating devices and an NT1. NT2 coordinates transmissions from a number of incoming links (user phone lines) and multiplexes them transmittable by an NT1.

ISDN PRI Frame Format: The B and D channels are multiplexed using **synchronous TDM** to create a PRI frame as shown in Fig. 9.7.

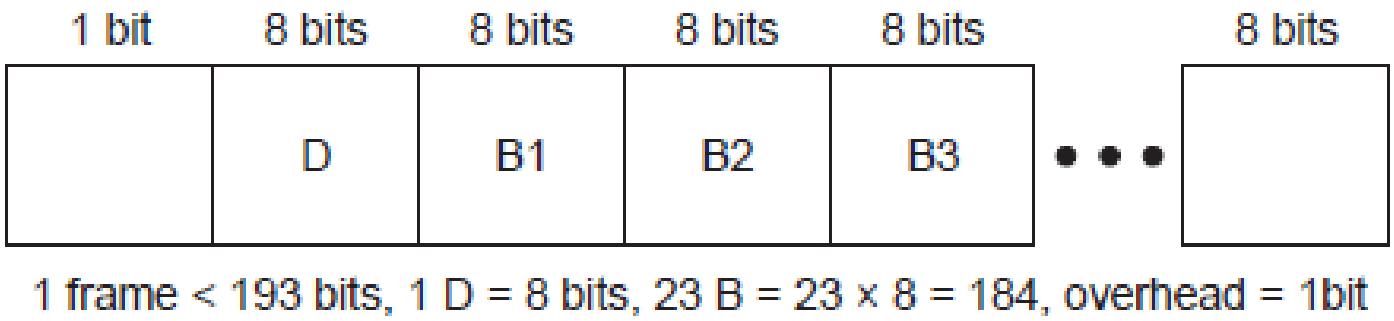


Fig. 9.7: PRI frame format.

PRI = 8000 frames/sec = 8000 × 193 bits/frame = 1.544 mbps.

Integrated Services Digital Network

Summary

N-ISDN — Carry data rating up to 64 kbps, ranging up to T1 rates. Sometimes used to refer to regular telephone and non-video capable systems.

B-ISDN — The communication standards being developed by the ITU to handle the high bandwidth applications such as video. B-ISDN will use ATM technology over SONET based transmission units to provide data rates of 155 Mbps to 622 Mbps and beyond.

Integrated Services Digital Network

Advantages of ISDN

- **High speed service:**
  - Before ISDN, normal phone carries only 2.4 kbps. ISDN digital phone lines can carry 128 kbps over the same wire.
  - ISDN can transfer two times faster than a 56 K modem.
  - The ISDN call setup (connect) is faster (2 to 4 sec) than analog devices (15 to 30 sec).
- **Cost advantage:** Cost is reduced due to -
  - Reduced retransmission of information
  - Fast information transfer
  - Simplified network management and maintenance
  - Reduced infrastructure and maintenance costs by offering multiple services through a single network

Integrated Services Digital Network

Advantages of ISDN

❑ High quality transmission

- Due to digital technology, less vulnerable to static and noise than analog transmission
- Transmission is highly reliable
- Voice conversation over ISDN also crystal clear, having the sound quality of an audio CD

❑ Simultaneous transmission

- Two B channels for voice, circuit or packet conversations
- One D channel to carry signals between your equipment and the phone company.
- Performs simultaneous functions
- Anyone can send a fax, receive a fax or utilize a 56 kbps internet connection while talking

Integrated Services Digital Network

Advantages of ISDN

❑ Multiple device connection

- As ISDN lines are divided into logical channels, up to eight devices (fax, telephone, computer etc.) can be connected on a single Basic Rate ISDN in any combination
- This reduces the additional wiring

❑ Conferencing

- As eight devices could be in use simultaneously, this may result in multiple call appearances.
- Allows to handle several calls at once or conference them together with one number

❑ Clear and quieter voice

- ISDN provides clear, quieter voice telephone service and easy to use call control features. Its caller identification features can screen incoming calls.

Call Management Features

Advantages of ISDN

❑ Call management features

- Call forwarding: Forwards call to a preselected number
- Call pickup: Call can be picked at another phone or station
- Directed call pickup: Calls from specific extension be automatically forwarded to a second number
- Message Waiting indicator
- Directed Dial: Incoming calls can be automatically forwarded to a central office, car, etc.
- Ringing options: Allows the user to use a distinctive ring to a particular call.
- Additional Call Offering.

Integrated Services Digital Network

BTCL provides the following Value Added Services (VAS) over ISDN:
Call barring, Abbreviated dialing, Call Conference, Call waiting, Wake up call (Alarm), Subscriber absence message facilities, Call establish facilities to busy subscriber, Hotline facilities, Call Forwarding, Temporary disconnection on request, Don't disturb message, etc.

Advantages of ISDN: Summary

1. High speed service
2. Cost advantage
3. High quality transmission
4. Simultaneous transmission
5. Multiple device connection
6. Conferencing
7. Clear and quieter voice
8. Call management features

Integrated Services Digital Network

ISDN Scenario in India

ISDN services were thrown open to the DOT subscribers in 1996. The DOT provides ISDN service on demand basis without waiting. This is the only value-added service remaining with DOT, while all other value-added services are given to private operators. The DOT has approved different customer premises equipment (CPE) that any subscriber can use in his ISDN line.

The only customer premises equipment provided by DOT is the network terminal NT. NT performs network termination functions, communication physical layer connectivity, power feeding, multiplexing and interfacing functions with one U-interface port and one S interface port. All other equipment and interfaces are provided to the subscriber according to his requirement.

Integrated Services Digital Network

Summary

ISDN Access

- Two levels of digital access by N-ISDN:
  - Basic Rate access: (2B + D)
  - Primary Rate access: (23B+D) or (30B+D)

ISDN Types

- There are two types of ISDN:
  - Narrow Band ISDN (N-ISDN)
  - Broadband ISDN (B-ISDN)

Broadband ISDN (B-ISDN)

B-ISDN provides the needs (High speed and large data handling) of the next generation technology. B-ISDN is a digital service with speed above 1.544 Mbps. The original ISDN is called narrow band ISDN (N-ISDN).

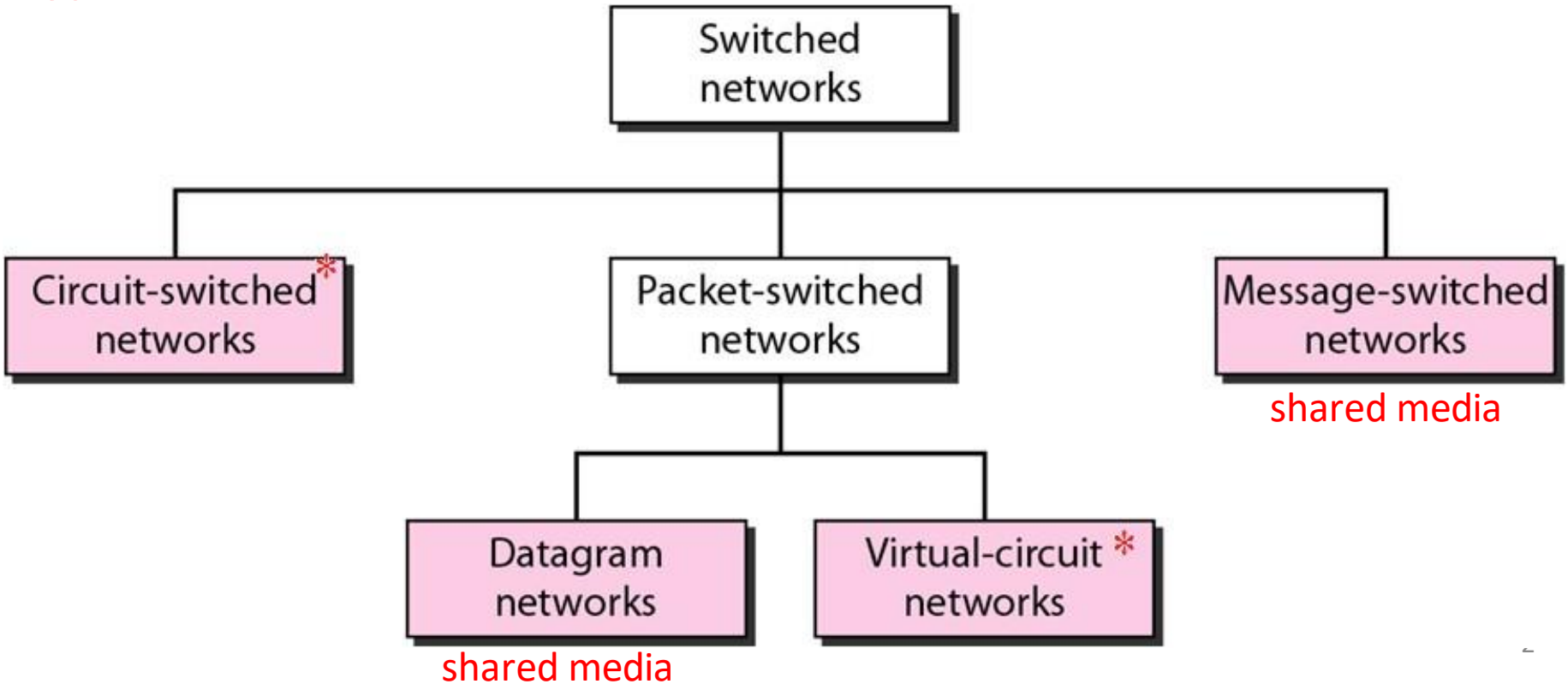
B-ISDN and ATM Switching

B- ISDN and ATM Switching

Broadband ISDN is defined as a network capable of supporting data rates **greater than** the **primary rate** (**1.544** or **2.048**Mbps) supported by **ISDN**.

The main aim of **B-ISDN** is to support **video** and **image services**.

Types of Switches



Shared and Switched Media

ATM Switches

The aggregate **bandwidth increases** as **more hosts** are **added** to the **network**.

ATM is based on a network of **interconnected switches** and **hosts**. These switches and hosts are arranged to form an **interconnected star**.

Since the **ATM networks** are based on **star topology**, **the polling access method** is used. An **ATM switch acts** as the **central hub** of the **star** as well as the **channel access administrator** or the **master device**.

ATM History

Asynchronous **T**ransfer **M**ode or **ATM** is a technology that has its history in the development of **broadband ISDN** in the **1970s** and **1980s**.

Introduction to ATM

ATM History

The **ATM forum** writes specifications and definitions for ATM technology. **ITU** approves this specification. **ITU-T** and **ATM forum** work **closely**.

*In **packet switching**, **delays** are **critical,** but **error rates** can be **sacrificed** for **speech transmission**. On the other hand, for **data transmission**, even **low error rate** is **critical,** and **delays** can be **sacrificed**.*

*These **conflicting requirements** have led to the **adoption** of a **broadband packet switching service** based on **A**synchronous **T**ransfer **M**ode (**ATM**).*

Introduction to ATM

- **ATM** = **A**synchronous **T**ransfer **M**ode
- The **standards** for ATM were first developed in the **mid 1980s**.

Introduction to ATM

Introduction to ATM

ATM is **originally** the **transfer mode** for implementing **Broadband ISDN (B-ISDN)** developed jointly by **ITU** and **ATM Forum**, but it is ***also implemented*** *in* ***non-ISDN environments***, where very **high data rates** are **required**.

ATM is a **connection-oriented**, **high-speed**, **low-delay switching**, **multiplexing** and **transmission technology** that **uses short** and **fixed-size packets** (**53** bytes = **5**-byte **header** and a **48**-byte **payload**), called **cells**, to **carry different** types of **traffic** including **voice**, **video** and **data**.

ATM technology has been **deployed** very **aggressively** within both **telephone networks** and the **Internet backbones**. However, it has been **unsuccessful** in **extending itself** all the way to **desktop PCs** and **workstations**.

Introduction to ATM

Introduction to ATM

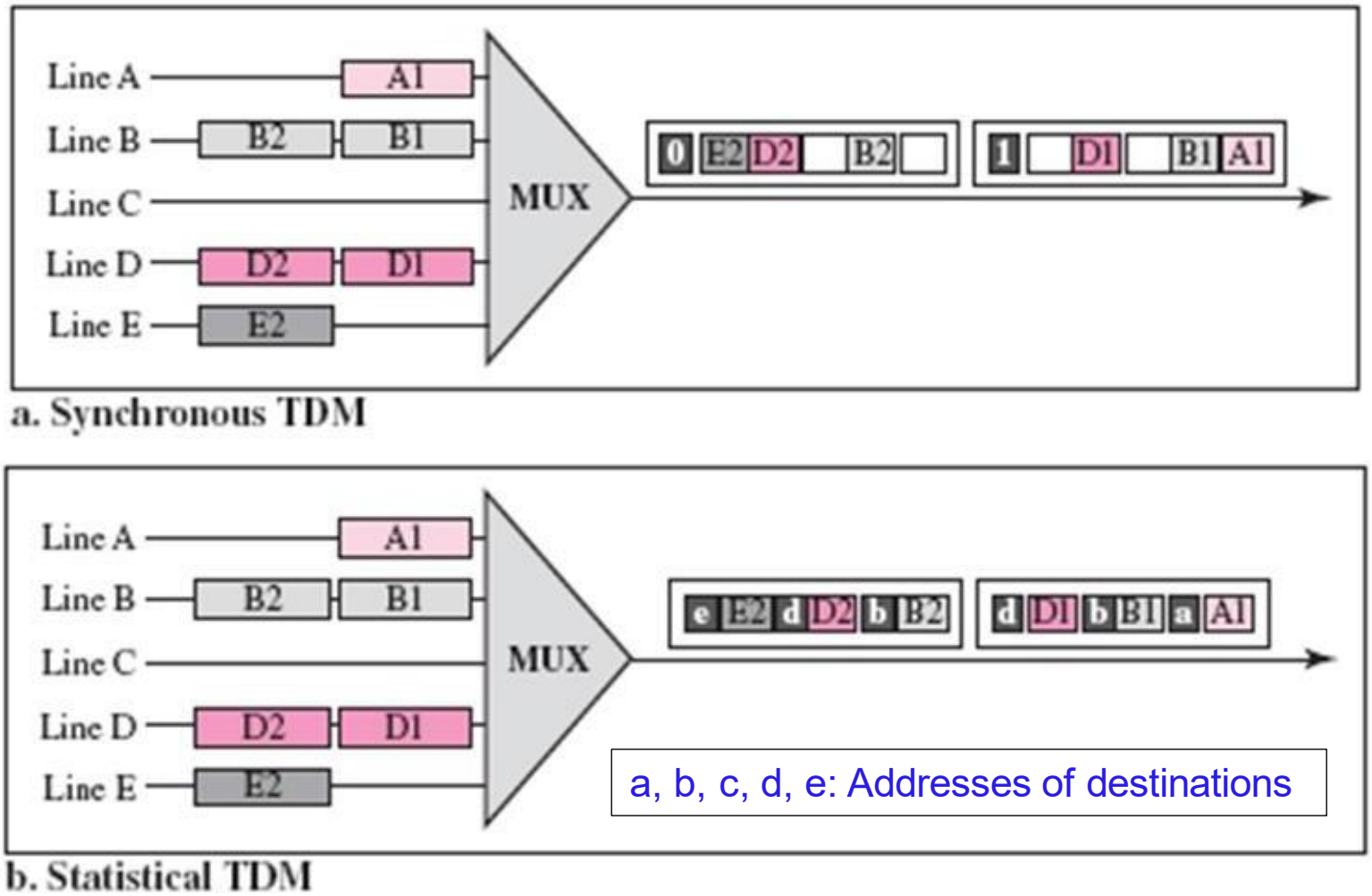
It is the next generation of networking technology to be used on the information superhighway. It is well **suited** for **bursty traffic** and allows **communications** between **devices** that **operate** at **different speeds**.

It **provides high QoS**.

ATM is a **high performance**, **cell-oriented switching** and **multiplexing technology** that utilizes **fixed length packets** to carry **different types** of **traffic**.

It is a **high speed packet switching technology** for the **broadband** ISDN (**B-ISDN**), in which **various** kinds of **communication services** such as **video**, **voice**, **text**, **data**- all are transferred over **common high speed links.**

ATM from Asynchronous TDM



ATM uses asynchronous/statistical time-division multiplexing (TDM) to **multiplex cells** coming from **different channels**. This is why it is called **A**synchronous **T**ransfer **M**ode (**ATM**).

Asynchronous Transfer Mode (ATM)

Comments

ATM *does* ***not imply*** *that* ***transmission links*** *or* ***switches operate asynchronously***.

The word ***'asynchronous'*** refers to the ***asynchronous operation*** *of the* ***clocks*** *at the* ***sending*** *and* ***receiving terminals***.
Any timing difference between **them** is **accommodated** by **inserting** or **removing empty packets** in the **bit stream** between them.
Optical fibres ***with their*** ***higher transmission capabilities*** ***and*** ***lower error rates*** ***are*** ***used*** ***in*** ***ATM networks.***

Advantages of ATM

The most **important advantages** or **benefits** of the **ATM** are:

1. **A** much **wider array** of **information** can be transmitted using **ATM technology**, e.g., **voice**, **data**, **images**, **CATV scans**, **MRI images** and **video conferencing**.

Advantages of ATM

Advantages of ATM (continued)

2. ATM delivers bandwidth on demand and is not dependent on applications.
3. Works at a data rate from 1.5 Mbps to 2 Gbps.
4. All types of networking, from LANs to WANs and from backbone to desktop can be integrated by ATM technology.
5. The service is connection oriented, with data transferred over a virtual circuit.
6. ATM switches use statistical multiplexing (asynchronous).
7. Higher Quality of Service (QoS).
8. Accepts variety of transmission media such as optical fiber or twisted pair cable.
9. Works with current LAN and WAN technologies and supports current protocols such as TCP/IP.

Concepts of ATM

Concepts of ATM

- ATM is a **connected oriented service** which is **implemented** by **using** a **virtual circuit.** In a **connection oriented service**, the **data stream** from **origin** to **destination** follows the **same path**.
- **A virtual circuit** is a type of **packet switching technique** which operates on the same concept as a **datagram** network; but, **routing** of packet is **specified before transmission**.
- In a virtual circuit, **data stream** from **origin** to **destination follows** the **same path**. Thus, **cells** in the **same connection** reach the destination in the **order** they are **sent**.
- It **eliminates** the **need** for **sequencing numbers** and **buffering packets**.
- **Data** from **different connections** is **distinguished** by means of virtual path identifier (**VPI**) and virtual channel identifier (**VCI**).

Concepts of ATM

Concepts of ATM (continued)

- **VPI** and **VCI** are called **connection identifiers**.
- **Here, each cell incurs** an **overhead** corresponding to the length of **VPI**/**VCI** which is **less than** the **length** of **source**/**destination address** needed.
- **No error control** or **flow control** is **needed** on a **link-by-link basis**. **End-to-end error control** is **provided** by the **terminal equipment**.

Features of ATM

Main Features of ATM

The **main features of ATM** are:

1. Packets are of **fixed length** of **53 bytes**, known as **cells**.

2. ATM operates in **virtual circuit mode**, i.e. **as if** there is a **dedicated connection**. If the necessary resources for a call are available, they are **reserved** for it. If **not**, the **call** is **rejected**.

3. No error or **flow control** is present on a **link-by-link basis**. **End-to-end** error control is provided by the **terminal equipment**.

4. Header is **short**, only by **5 bytes**, and **identifies** the **virtual connection** with **error control bits**. No **sequence numbering** is provided.

5. Information field is **short** only **48 bytes**. Short cells **reduce** packetizing **delay** at the **sending terminals** and queuing **delay** at **switching nodes**. (On the other hand, **packet switching** uses **256 bytes** in **each packet**).

ATM and SDH

Main Features of ATM (continued)

6. A **policing function** is provided which enables the network to **ensure** that the **user** is **not sending** and **receiving information** at a **greater rate** than **what** he has **paid for**, i.e. different service categories.

In **ATM**, information is formatted into **fixed-length cells** consisting of 5 bytes of **cell header** and 48 bytes of **user data** (**payload**). That means **users** send all kinds of **information** as a series of a 5-byte **header** and 48-byte **payload**.

Synchronous Digital Hierarchy (SDH)

The function of **S**ynchronous **D**igital **H**ierarchy (**SDH**) **transmission nodes** is classified into **path multiplexing / demultiplexing function**, **cross-connect function** and **add-drop function**.

All **these functions** are **flexible** between **low** and **high bit rate** circuits.

B-ISDN/ATM Network

ATM Network

B-ISDN involves **both ATM** and **SDH** technology. Fig. 10.1 shows a typical BISDN/ATM network.

Fig. 10.1: B-ISDN/ATM Network.

Types of ATM Connection

Types of ATM Connection

ATM is a **connected-oriented service implemented** by **virtual circuit switching**. The following **two types** of **connection** are used in **ATM**.

1. Permanent Virtual Connection (PVC)

PVC is a connection that is **setup** and **taken down manually** by a **Network Manager**. A **set** of **network switches** between the **ATM source** and **destination** are **programmed** with **predefined values** for **VCI/VPI**.

Drawbacks of PVC connections: **(a)** They are **costly** because two parties pay for the connection all the time even when it is not in use. **(b)** A connection is created from one source to one single destination. If a source needs connections with several destinations, it **needs** a **PVC** for **each connection**.

Types of ATM Connection

Types of ATM Connection

2. Switched Virtual Circuit (SVC)

SVC is a connection that is **setup automatically** by a **signaling protocol**. SVC creates a temporary, short connection that exists only when data are being transferred between source and destination.

An **SVC** requires establishing and terminating phases. **SVC** is more **widely used** because it does **not** require **manual setup**. However, it is **not reliable**.

ATM Network/Switch Interfaces

An **ATM network** consists of **switches** and **end users** with necessary **interfaces**. It uses **two types** of **network interface:**

- User to Network Interface (**UNI**)
- Network to Network Interface (**NNI**)

ATM Network Interfaces

ATM Network/Switch Interfaces

ATM Networks/switches offer **two types** of **interfaces**. They are:

1. Switch to User Interface or User to Network Interface (UNI): **UNI** is the **standard technical specification** allowing **ATM customer equipment** (**CEQ**) from various manufacturers to communicate over a network provided by yet another manufacturer. It is the interface employed between **ATM customer equipment** and (either) **ATM switch**.

2. Switch to Switch Interface or Network to Network Interface (NNI): It is an **interface between nodes** within **the network** or **between different sub-networks**.

Another interface called **ATM Inter Network Interface (INI)** is used for **intercommunication**. It is also used for **operational** and **administrative boundaries** between **interconnected networks**.

ATM Network Interfaces

ATM Network/Switch Interfaces

It is **based upon NNI** but **include more features** for **ensuring security**, **control** and **proper administration** of **inter-carrier connections**.

Fig. 10.9 illustrates a typical **ATM network** with the **interfaces**.

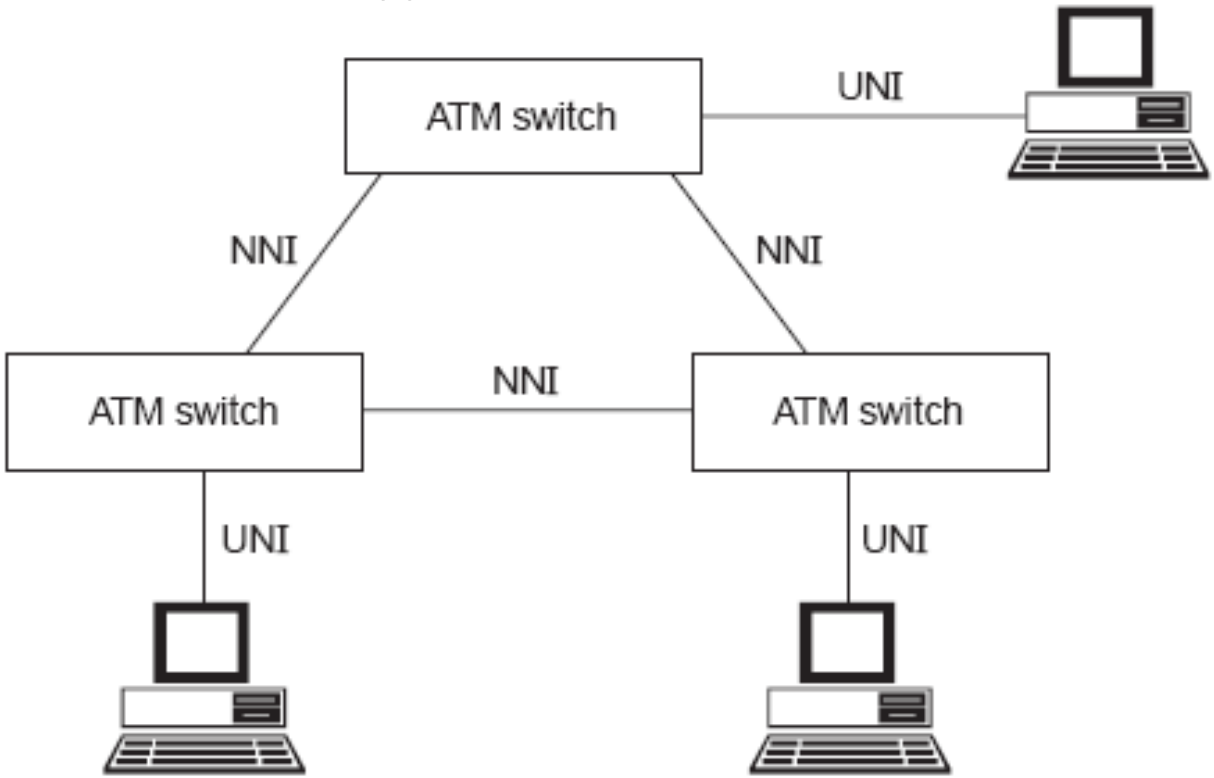


Fig. 10.9: ATM Network Interface.

ATM Multiplexing

ATM Multiplexing Technique

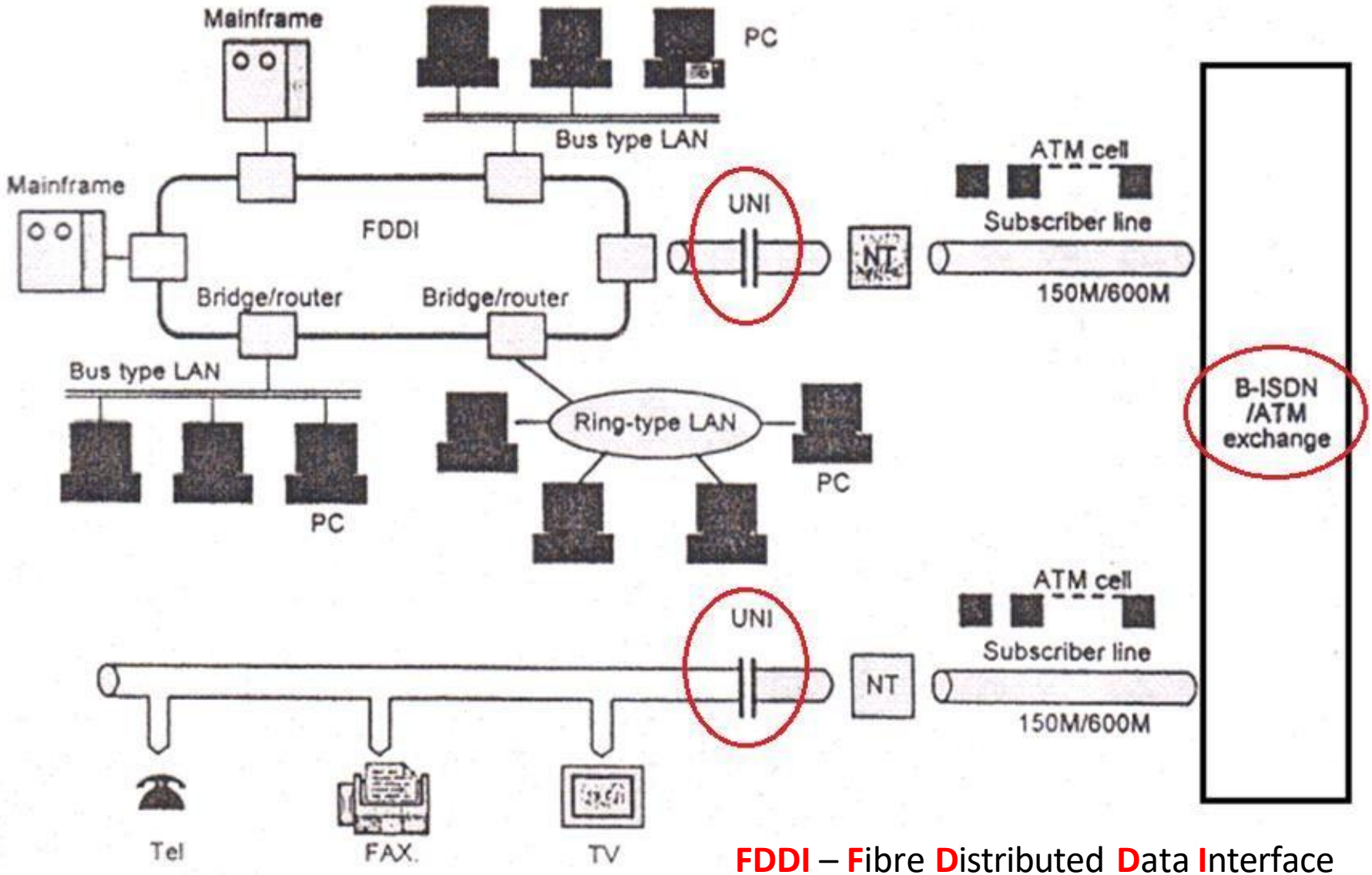
In ATM, **multiple** streams of traffic can be **multiplexed** (**TDM**) on **each physical facility** and can be **managed** so as to **send** the streams to **many different destinations**.

This **enables cost savings** through a **reduction** in the **number** of interfaces and facilities required to construct a network.

Fig. 10.2 shows the example of a user network interface in an **ATM** exchange.

UNI and ATM Exchange

User Network Interface in an ATM Exchange



FDDI – **F**ibre **D**istributed **D**ata **I**nterface

Fig. 10.2: The example of a **user network interface (UNI)** in an **ATM exchange**.